Edwin Arkel Rios Personal Page
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Architect of efficient deep learning systems, specializing in model compression and HW-SW co-design.

Education

National Yang Ming Chiao Tung University Hsinchu, Taiwan
PhD in Electrical Engineering and Computer Science, Cumulative GPA: 4.24/4.30 *September 2019 - Present*

National Cheng Kung University Tainan, Taiwan
Bachelor of Science in Energy Engineering, Cumulative GPA: 3.76/4.30 *Sept. 2015 - June 2019*

Experience

Tranxform AI, Generative AI R&D Intern *December 2025 - September 2026*

- Extended HQQ with per-layer bit widths for audio models, quantizing Whisper's decoder to 1.58 bits and its encoder to 3 bits.
- Built a quantized automatic speech recognition (ASR) framework in Hugging Face transformers (32 quantization configs from four PTQ libraries, 26+ models, 100+ datasets); the work produced 5 papers (3 accepted).

NYCU PCS Lab & NTHU MIS Lab, Research Assistant *July 2019 - Present*

- Developed methods for knowledge distillation, pruning, and efficient training of vision models, leading to 10 first-author papers in international venues.
- Led and mentored 18 students, resulting in 14 publications and 3 awards.

Selected Research Projects

Low-Bit Quantization of Large Audio Language Models *2025 - 2026*

- Benchmarked 18 ASR and audio language models across 25 languages at 3 to 8 bits: the quantization penalty falls as Whisper grows and does not depend on a language's pretraining data.

Input Pruning for Acceleration of Transformers via HW-SW Co-Design *2023 - Present*

- Designed a module that keeps as few as 10% of input tokens at accuracy on par with SOTA.
- Proposed a methodology to select token reduction settings for an FPGA accelerator.

Parameter-Efficient Fine-Tuning (PEFT) of Foundation Models *2023 - Present*

- Developed a lightweight adapter: at least 6.8% higher average accuracy than other parameter-efficient methods, with at least 123x fewer trainable parameters than SOTA.

Efficient Discriminative Feature Selection for Fine-Grained Image Recognition (FGIR) *2021 - 2024*

- Engineered a novel mechanism for selecting high importance regions for fine-grained classification at a much lower computational cost.

Skills

Languages: Spanish (native), English (fluent), Mandarin Chinese (professional working proficiency)

Programming Languages: Python, MATLAB, C++, Bash

Libraries: PyTorch, Hugging Face transformers, HQQ, bitsandbytes, scikit-learn, OpenCV, NumPy, pandas

Tools: Git, Docker, Linux, Weights & Biases, Vitis HLS for Xilinx FPGAs, Jupyter Notebooks, LaTeX, HTML

Awards and Certifications

Merit Paper Award, 39th IPPR Conference on CV, Graphics and Image Processing (CVGIP) *2026*

Excellent Paper Award, 38th IPPR Conference on CV, Graphics and Image Processing (CVGIP) *2025*

Associate Fellow (AFHEA) of AdvanceHE, UK Professional Standards Framework in Higher Education *2025*

NCTU Academic Achievement Award during R.O.C. Academic Semester 108-1 *2020*

NCKU Outstanding Student during R.O.C. Academic Year 105 and 106 *2018*

Selected Research Works

2 journal papers, 14 international conferences and workshops, 5 local conferences; 6 under review.

How to Choose Your Teacher for Fine Grained Image Recognition *CVPR W 26*

Cross-Layer Cache Aggregation for Token Reduction in Ultra-Fine-Grained Image Recognition *ICASSP 25*